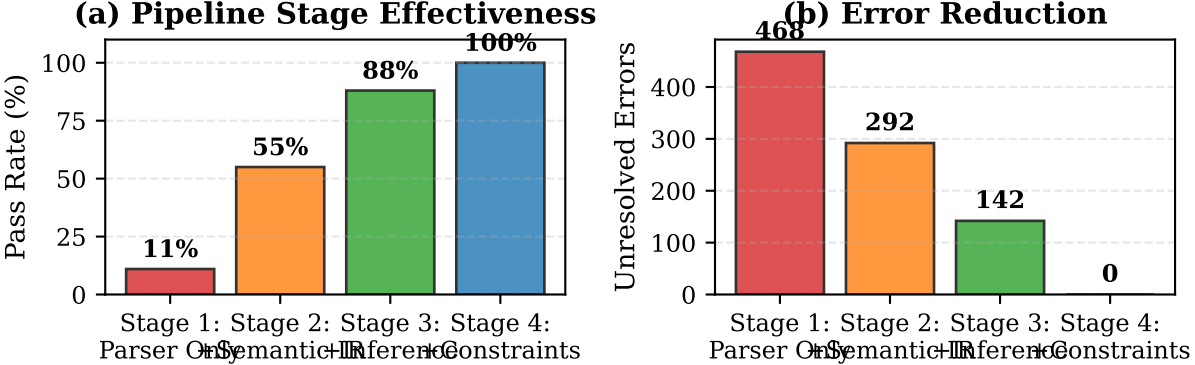
(a) Pipeline Stage Effectiveness
Pass Rate (%)
0
25
50
75
100
11%
55%
88%
100%
Stage 1: Parser Only
Stage 2: +Semantic IR
Stage 3: +Inference
Stage 4: +Constraints
(b) Error Reduction
Unresolved Errors
0
100
200
300
400
468
292
142
0
Stage 1: Parser Only
Stage 2: +Semantic IR
Stage 3: +Inference
Stage 4: +Constraints